

NATURAL SELECTION

Survival of the fittest is about an organism being better suited to a particular environment and having an increased chance of surviving long enough to pass its genes into the next generation.

The Theory of Evolution includes the suggestion that the mechanism for evolution was natural selection.

The three key terms to understand this idea are ***variation***, ***competition*** and ***selection***.

Variation

Individuals that possess a favourable variation (or phenotype) will have an increased chance of reproducing and passing on this variation (through their genes) to their offspring.

Inheritance of this variation (or phenotype) will increase the chances of survival of the offspring and hence the possibility that they will contribute their genes into the next generation.

Over time and many generations, if this variation continues to provide a selective advantage, the number of individuals within a population that show the favourable variation will increase.

Competition

Individuals within a population compete with each other for resources such as food, shelter or mates. Those with a selective advantage over other individuals are better able to compete for the resource.

Selection

Organisms live within ecosystems, which are made up of ***various living (biotic) and non-living (abiotic) factors***. These factors contribute to selecting which variations provide the individual with an increased chance of surviving.

It is for this reason that these factors may be referred to as ***selective pressures*** or ***selective agents***.

Biotic Agents - predators, disease, competition for food (animals), competition for nutrients and/or light (plants), mating partners.

Abiotic Agents - heat and cold, wind, availability of O₂ and H₂O, pH (acidity) of soil and water, availability of space, shelter, nutrients.

The following diagram summarises the factors that affect natural selection.

